6.3 Ellipses

Name _____ Period ____ Date _____

Pre-Calculus 6.3 · Student sheet

I can…

- write the standard equation of an ellipse.
- find the center, vertices, and foci of an ellipse.
- find and interpret eccentricity.

Vocabulary · fill in the missing word

Ellipse — the set of points whose _____ from two fixed points (the foci) have a constant sum

Major axis — the chord _____ the foci; its endpoints are the vertices

Minor axis — the chord through the center _____ to the major axis; its endpoints are the co-vertices

Eccentricity — $e = \frac{c}{a}$, which measures how _____ an ellipse is

Part 1 · String and tacks

Draw an ellipse with the tacks and string, then move the tacks and draw again.

Tacks	Shape: rounder or flatter?	Sum of distances from one point to both tacks
Far apart		
Close together		

Part 2 · Notes and examples (fill in each note, then work the example)

Standard form (center (h, k)): horizontal major axis $(x-h)^2/a^2 + (y-k)^2/b^2 = 1$; vertical major axis $(x-h)^2/b^2 + (y-k)^2/a^2 = 1$, with a _____ b.

Example 1

Write the standard equation of the ellipse with center (0, 0), vertices (±5, 0), and co-vertices (0, ±3).

Foci: c units from the center along the major axis, with c^2 = _____.

Example 2

Find the foci of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

Example 3

Write $x^2 + 4y^2 + 6x - 8y + 9 = 0$ in standard form. Find the center, vertices, and foci.

Example 4

Write the equation of the ellipse with foci $(0, \pm 2)$ and vertices $(0, \pm 4)$.

Eccentricity e = c/a: near 0, the ellipse is nearly a _____; near 1, it is long and thin.

Example 5

Find the eccentricity of $\frac{x^2}{25} + \frac{y^2}{9} = 1$. Is it nearly circular?

Part 3 · Partner practice

Partner 1

Find the center, vertices, and foci of $\frac{x^2}{9} + \frac{y^2}{25} = 1$.

Partner 2

Write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form. Find the center.